ActInf Textbook Group ~ Cohort 4 ~ Meeting 21 (Chapter 10 part 1)

TextbookGroup/ParrPezzuloFriston2022/Cohort_4 | Cohort 4 ~ Meeting 21 (Chapter 10 part 1) | 1:00:54

all right welcome
everybody
it's November 14th and we're in cohor 4
first discussion on chapter 10
so who wants to begin with some thought
or question or any reflection about
10 any can raise their hand or just go
for it but what they thought 10 was
going to be or some component of what
they remembered of
it or just some quote that we could
start with or question that they already
wrote sure uh I mean just kind of at a
cursory glance at this chapter it's just
they're
basically going through a variety of of
other fields that may or may not be
related to active inference and kind of
comparing and
contrasting uh I think it helps to
understand what kind
of
um kind of trend or or or body of
knowledge in which you know active
inferences has its roots or where it's
starting to assert itself as a as a
distinct field um it's interesting
learning a little bit more about for
example uh the helm Holden view of like
um unconscious inference as being like
one of the key underpinnings of um
active inference here um seeing more on
the Basi and brain hypothesis and
reinforcement learning and other fields
that we see mentioned through the

textbook but it's here that we get like
a direct kind of comparison um and in
addition to that I think one of the key
overarching themes that they're trying
to make is that active
inference combines a variety of kind of
traditionally separate or separated um
cognitive functions like perception
action
planning attention and and so on that's
the the title of the chapter is active
inference is a unified theory um makes a
lot of
sense nice very
nice so I
um um there's something about the
textbook overall um that um I found
myself wondering about when I was
reading chapter 10
um which is
uh that there are many places in the
book where it's not really
clear if we're talking only about how
the human brain functions versus how
cognitive systems
function and so there are places
where
um where it's almost uh it's almost it
feels like a a tease like let's let's
explore this is not just about the human
brain this is about other cognitive
systems but I think for people who uh
the
uninitiated and of course I'm one of
those people
at the moment but um it's it's tough
right because I've created certain
expectations about looking for how does

this generalize to all kinds of
cognitive systems but then in the last
chapter the focus is and the wording is
used for the rest of the chapter it's
almost like we're going to introduce
these Concepts like you would for a
psychology
textbook um and so they you know those
various uh terms that are used in
Psychology are addressed and how do we
think about them from an active INF
point of view then the section on social
and cultural
mechanisms I think needs to be either
deleted
or completely
rethought
um so when I kept going back to the
figure in um chapter one I think it's
figure
1.2 where it has a high road and a low
road and in the center it has active
inference and it has autop poesis and
nich
construction well Auto Poes and inch
construction are brought up very
infrequently throughout the
textbook and I think there's an
opportunity for chapter
10 to to go back to other places and
each
construction
and and there are articles out there
that I'm finding
fascinating uh on the wording I'm still
learning new words but what is this one
called um joint agent I love this this
article

by ruinberg I don't know if this is a
one that's um cited in in the book it
might be uh but um joint agent
environment
systems and I think that that's the
place where really cool stuff can be
laid out uh for people who want to take
this FR Market into the social sciences
and
anthropology
thank
you any other like thoughts on 10 or
overall whether it's someone's like
first time at 10
or
repeating
well I had I have another thought and
kind I'm curious as to how
others you know feel about this
observation I don't
know yet how where I would suggest to
include this somewhere in the book
but um if you know if most of the
information that's presented in the book
is backed up by data that's been
gathered about the human brain then I
think it would be interesting to point
out that the human brain is actually a
very new centralized nervous system in
the history of living
organisms and it's probably between
100,000
35,000 years old uh meaning that the act
the present day uh the modern human
brain
organization presumably reached that
point about that around that time
100,000 to 35,000 years ago so what I

think is super interesting is
that the brain the human brain that such
a recent development has unveiled so
much interesting information
about um the ways in which we navigate
the world
and uh as sentient systems whatever that
means right and said that that needs to
be I think very clearly stated somewhere
in the book what do we mean by a
sentence
system um but anyway I just thought I
would bring that up and ask you uh other
is like what do you think of that
observation anyone can go or I'll give a
thought Andrew go for it sure I I just I
want to see if I understand the the
question properly but I mean
for me I think you know for
example looking
at the human brain and neurobiology
looking at
cortical micro circuitry for example
finding a relationship between that
and message passing the kind of things
we find in like chapter five in the book
I mean I think it's
very I think it's very interesting I
think it's it's fruitful and
um I think that there have been
empirical studies that are being done um
and of course one of the biggest tricks
is like to see like how like given we're
external observers um we have to kind of
see how the data
behaves and figure out what is a
reasonable proxy in the data for what's
actually going on in the brain um so um

I mean I I find a lot of validity in at
least attempting
to kind of make this connection between
like the the brain however new it is um
and and maybe not just the brain itself
in terms of its circuitry but also like
inferential processes trying to
understand how those function in the
brain or applying that as a kind of
metaphor or analogy for looking at
potenti
uh entire Society although that's an
area of that's the kind of work and
active inference that I'm less familiar
with so I don't want to overp speak on
that um I I don't know if that was like
a like a um you know onetoone like
relevant response to your question so
I'm just yeah I'm happy to keep talking
about it if I understand
you yeah I guess uh so I guess what I'm
trying to uh appreciate
like I don't really have
anything super compelling to say other
than to note or make the observation as
context for this book I think is that
the first centralized n uh nervous
systems showed up in the evolutionary
record I think about 570 million years
ago and and so and the the book is about
sentient cognitive you know s sentient
systems
presumably and so I had to do some
digging and think well are we talking
about uh centralized nervous systems
that evolved emerged 570 million years
ago and then you know from that
perspective then I would uh I Marvel at

the fact that you know humans have this
incredible
engine
that evolved very
recently and yet it has the capacity to
not just do everything that's in you
know I I've completely sold and believe
in the whole active inference framework
at this point you know I just think it's
it's just revealing so much that I can
know information about
but I find it super interesting that
this brain can
actually externalize our generative
models and that we can look at those
generative models and that's why we we
can do science right um and so it's not
so much that I I think it makes the work
less
valid because we are such a recent
development in human in evolutionary
history but rather
how this these
processes phenomenally led to this brain
that can do not only what you know being
able to contemplate and think about our
generative models but to actually create
generative models through artificial
intelligence and that is something about
this book that's an Insight that can
come out as one
of a really important aspect of the book
somewhere so that that's what I'm trying
to
convey thank you yeah these are great
gray
points here's a figure we often return
to in this path integrals paper so it

was like absent literature until it was
drawn showing the continuity of the
basil cognition and the different types
of
things and so yeah there's that
[Music]
[Music]
some fundamental aspects to that
reflexivity there's a lot of it brings
everything you know it it it opens up a
lot which is why and and so I think this
is a really important point that you're
bringing up like do we approach the book
is what it is 2022 textbook can never be
changed they could release a different
version we could generate a massive
amount of educational material and
develop ecosystem around this open
source textbook Etc so so it's not about
changing from how I see it what that
book is but how it is approached and
conveyed could run the gamut
from an extension to a statistics
course ranging to including deeper
investigations like you brought up of
the longer Arc of
sentience and just allowing that to
open more and into
slower different regimes of
attention yes in in this framework
Daniel where does con context come
into like in a in a formal
way what does anyone think where does
context come into
play I'm S I want to make sure I
understand the question again
uh context in the form of do you mean as
far as like active inference and and

behavior goes or do you mean the field
itself like it's
context um okay
um I mean I I'm let's go back to the
figure
1.2 where you know it's the high road
and the low road where in
this in this
world doeses context come into
this
science there's probably a ton of
answers but one short narrow answer is
self-organization or just broadly the
partitioning or differentiation that
allows us to say self versus nonself or
just any kind of continuation that's the
context that's the niche so the first
particular partition that enables us
from not being able to say anything
about that thing that thing not
existing to it being a thing it comes
into
description um as
contextualized another related point is
this as if instrumentalist view that the
the it doesn't take
on um needing to provide an account of
how whatever context is applies to the
rat in the
maze
so that's why it centers
potentially that self-reflexivity of the
Observer because the model not to say
it's boring not to say it's simple but
it's just an artifact resulting from a
different
process yes um so so
I okay

but when you know when you're talking
about something
like how do I want to ask this question
um so you use the word
artifact I'm just going to stick for now
with the idea that that artifact in
figure
1.2 is applicable
across all sorts of contexts across all
kinds of agents and I think what what I
would like to know is where where is the
I know I shouldn't use that word but
where's the boundary I was going to say
Mark of blanket I'm not going to use it
but where's the boundary um in other
words in what systems does this arti
fact not is is it not
applicable yeah very common question
about the applicability or the scope of
this type of
modeling and and so if it's left us as
as an open question that's okay and I
won't you know push any further but I
it's really
been does anyone want to give a thought
yeah I I can give a thought and I'm uh
in cohort five so I'm just uh I guess
piggybacking off this um
but um I think that one of the you know
active inference is a model of
understanding really any natural system
that you want to look at
um that behaves in under the certain
Dynamics
um it could be applied to just about
anything so the context is really going
to be up to the person using it so you
could have any context and you know try

to draw the boundaries anywhere and um
how you draw those boundaries is going
to determine what the organism or
particle that you're looking at is and
what the environment is so I I think
that's one of the cool things about
active inference is it's so broadly
applicable and where you draw those
boundaries makes all the difference in
how the models turn
out thank
you that was really help one note on
this is figure 1.2 is not a formal
argument it's more evocative so these
different terms might apply in different
ways to to different systems like
planning as inference not all systems
plan and then even again pulling back to
the instrumentalist layer you don't have
to model the system as planning but
maybe you do model even an inment system
as doing some kind of extended planning
or vice versa so they're not requisite
they're they're just uh some train stops
on the low road and the high road and I
think that is a layer that that does
kind of make sense to pull back to not
that this is exhaustive or um a Panacea
but it's just a framing that helps
organize the the book and the work and
also situates active inference at the
intersection between actually building
something from the ground like what
dennet might call the cranes approach
and the tethering or the association to
the high road principle the Skyhook in
that same kind of denit metaphor so that
there's something that can be

realized with real software tools and
real relationships between
the math and the Theory and the
implementations finding new
useful through and
throughs
and therefore having nothing to say
about any given Thing by virtue of being
applicable to any modeling
situation
Brahman uh yes Daniel I was just
um the figure 1. two I I I view it as
you as you said sort of a you not not a
fixed thing not really a a a pathway
thing but but in ways a lot of those
things were historical developments in
the way people scientists and
philosophers were
considering um the progression of
understanding of the sentient being or
the we use human being in its
environmental Niche so that sort of
constant question of how is it that we
persist how is it that we survive what
is it we do how do we adapt and I keep
going back and I was talking to you
before about
10.3 and Daniel dennett's comment in
1978 about the necessity to to Really
model the whole of Guana which is really
about how do we understand the human
being as a as a
whole and so there's a lot of I mean and
I also there's a distinction I think
between this conversation on modeling
something and a conversation
on um understanding the Adaptive Dynamic
nature of an evolving creature in its

environmental Niche so in a way I sort
of separate out those things was a
modeling sort of the acting on the
consideration of how it is that we
continue to survive or not survive
whatever it might
be
so I mean one thing I think that I've
learned in my travels through active
inference is its connection historically
to a lot of thought from a lot of people
and a lot of um consideration of of what
we do how we do it and that's
including
our our our sort of evolving ability to
be cognizant of ourselves and understand
how we are actually making decisions so
it's that
stepping back out of our precisions
actually um enabling us to fine-tune our
own basic mechanics so that we can
actually see ourselves clearer in our
environments so that we can then
function to survive and make better
choices and then we can begin to sort of
model that I suppose but I always have a
question I'm rambling on a bit but I
also I always have a question about
someone creating a model when they
actually have a biased view
of which is which is the nature of it a
biased view of what they're what they're
doing so I think it's
a
um the distinction needs to be made
between the actual uh what the fre fre
energy principle states which is we must
reduce or minimize uh free energy and

the actual process theory of active
inference in which we we bring that
into um pragmatic play in our lives as a
whole being in our Niche and then
wrapped in that is then I think we can
then subsume all of those other
questions that
arise whatever I just said
wow but no but anyway one point two is
that I mean if you go into each of those
different areas autop poesis um
predictive coding blah blah blah you
know they they've all got a sort of a
progressive historical evolutionary type
footing um on which um I think active
inference rests and um I even found
another person in America that I was
going to speak to about alre as I don't
know if you've ever heard of him but um
yeah there there's like a you know it's
as if as a as a whole being we're we're
evolving we're evolving to be able to
recognize ourselves and uh what we do in
our
Niche I'll stop
there yeah that's a very interesting
point evidence by
the perspectives people bring the skills
that are required the kind of
collaborations that would
entail skillful
usage a lot of different aspects to it
yeah yeah and that's sort of like one of
the follow-ups after the well the scope
question is it applicable to Any Given
system and it's like well the question
presupposes if it's a system that's that
someone's thinking of then yeah that's

the system just go ahead and describe it
so people try to hold that out like as
if it's in a rug pull to talk well it
could apply to everything it's like
yeah um and then from there the natural
question is like well then what does it
do and the whole iguana it's funny
because people make what they believe
are Justified true belief assertions
about human behavior and this molecule
does this and it's like not not to say
that smaller brains are simpler of
course I'm always bringing up the insect
brains here but think about how many
experiments go into determining the role
of like neurotransmitter at a certain
synapse and so it's like way less
evidence in humans obviously important
everyone cares but then to overstate the
the
Precision we're not even at the whole
iguana this is the whole chapter 7 tze
it's like that's that is whole in what
it is it's a pedagogical example and
then by iterating with a real animal in
the Maze then there's more and more and
and also becomes clearer and clearer how
the map's not the teritory from the
first draft to to the one millionth
draft map's not the territory under
whatever constraints the map and the
territory co-evolve
under so it's are the the questions that
that help make chapter
10 actually about connecting the dots
and about anything more than just a
literature review
on psychological and philosophical

topics
and actually
so when you you brought up the the word
bias right now would it be a correct
interpretation of generative models to
say that um I think active
inflence what
it's one of the implications that that
seems to make sense to
me is that that we are constantly in the
process of creating
biases uh because by creating biases we
um you could say that a bias is just a
model in our heads of the
world and in science we're constantly
coming up with these models and but then
we get a chance to discuss them and put
them at
risk um but
that but that's just kind of an example
of how powerful I think the power of
this perspective that it actually allows
me to think about something like bias
that we talk about all the time and not
just well it's just bias well what is
bias I'd like to read a little section
from a short piece that I wrote so
basically um active inference Learners
often look at representations such as
figure 4.3 from the 2022 textbook or
figure 1.2 or figure any of the
textbook seemingly expecting to find
sophisticated phenomena such as affect
or narrative reflexivity in or on that
representation two key aspects of my
response in that situation first if
active inference were to include even
relatively simple cognitive phenomena

learning or tension in its essential
particular formulation particular
partition this would explicitly restrict
the scope of analysis to systems with
that character and silently diminish our
Capac to distinguish across that
difference second natural language or
conceptual descriptors of cognitive
phenomena are secondary attributions
better understood as relational
rhetorical associations assertions using
a pattern language of phenomena about a
given realized generative model rather
than fundamental or intrinsic aspects of
the system of Interest
itself not to be verbatim but I think
it's a really important Point like bias
has a narot technical definition you'll
see in a statistics textbook it's what
biases the data and does anything
different than a pass through it's just
a feature of how discriminatory a
signaling algorithm is and then of
course there's broader senses and and
alignments with with with broader um
usage of those terms just like with so
many of the terms in the actin ontology
and I think it's relevant that
especially when talking about the
broader systems of Interest or the
phenomena that there there are not the
basil low road they're not the assertion
they're not what actually implements the
system they're descriptors of systems
with observers which is the the setting
where more can be done and and
it's a
better discussion

setting when it's
assertion based not
like not not to go too hard on this one
point but I think that that that puts
the space as us as generative model
interpreters in between us and any
created
artifact without that articulation we're
we're just not set up to have a
productive epistemic
conversation thought on that and I like
that quote that you showed Daniel about
the comment of the vagueness and
difficulty with philosophical questions
is intentional so that it's a
generalizable model we don't want to be
pigeon rolling it um but I I think bias
and this is my perspective on it is a
part of the learning process and if
you're learning anything that you hope
to be
generalizable then when it's accurate it
was useful information and when it's in
a situation where it's not accurate then
call it bias and anytime you're trying
to generalize something you are creating
a bias that you hope is going to be
helpful and useful at some future point
But as soon as it gets out of that
context where it's helpful and useful we
just call it bias it's skewing your
understanding in a certain
way yeah that's very important like if
someone says that was a biased decision
to think like how could they violate
this or that but if it was like no they
picked up on subtle cues it was a biased
decision

ision be like wow great for them in that
situation so what what is
really the
discussion not even to that topic but
just this is that that's stepping back
from even our words and our
symbols how does that really play out
were Learners on
that all all with each other in the
textbook group
there is a technical part to active but
I think conversations like this chapters
like this they show it's not simply that
way the whole low road could be ripped
out and replaced
potentially even in the the message
passing live stream right now with
Magnus kudal um like generalized free
energy generalizes variational free
energy and expected free
energy so those are some of the main
topics we talk about about like equation
2.5 and 2.6 in this
textbook but if those formalisms last
for several years to to decade then
they'll have
been long lived even
by things not accelerating
standards so there's just a lot of
interesting pieces like as the textbook
recedes
then
do do we continue to drive back to the
2022
textbook do people want spaces of more
open
straightforward I don't
know something in between all of the

above of
course um personally I need more time to
to think about
that I I'm still there's so much in the
book that I still I'm processing and
learning like that's kind of the funny
joke at the end of the
book making the the
extracurricular claim that active inform
is not something that can be learned
purely in
theory that's why it's such a great I I
love that about it
because honestly when when I as I
read I have to pause and and then step
away and think about how to S supply to
this system that system other stuff I'm
thinking about and then I come
back
um but you know I'll just share with you
guys uh that um so I've been studying
Buddhist psychology for
decades and what one of the things that
is really mindblowing to me is how the
very high level Buddhist
psychologist um
scriptures
um are so much have very much to do with
active inference yeah
so so yeah I mean I I guess I'm just
thinking about applications and then
going back to the ideas digging deep and
trying to master very complex Concepts
in this second
way Rin
um that that's can I can I just ask a
question um did you just say with the
Buddhist psychology that um that active

inference was congruent seems to make
sense of of a lot of the philosophy was
that what I
heard well I'll I'll give you a spe
specific example um so for ex so in the
um
so Buddhist psychology is very also very
applied it's uh very abstract and very
applied and so one of the applications
is meditation so what is meditation what
you do with your mind when you're
meditating and essentially what you're
doing is is uh putting at risk all your
generative
mod that the way that you find the idea
behind that you know it's it's called um
investigating your
mind it's not you know if you have a
horrible thought about killing somebody
you know it's not that you just say oh
no stop that thought it's examining why
do I have that
thought and then you you ask yourself
that but to me when I I was reading
about active inference is about yeah we
create this generative models like since
I'm an an apologist and I worked on
these these different cultures I mean I
I've lived in cultures where people will
say we men can kill women and not be
punished okay well that's a model that
these people carry in their minds right
and so in Buddhist
psychology it's what is the the default
kind of tape layer that's going on in
your mind that's driving everything that
you do that you never think
about and that's what it is that's it's

a simp it's very
simple and uh it's very consistent with
active
inance glad you think so yeah there's
like several articles and lines of
research with like attention meditation
Buddhist psychology different knowledge
Traditions so that's definitely very
interesting
bronwin um yeah well because the reason
I ask is is that that I am an Alexander
technique teacher which most people
wouldn't know what that was but the
Alexander technique has a a history but
it goes back in time to so 100 years ago
with uh John dwey who was an
educationalist philosopher in America
and um I think it was stated once on on
this book club that you know Jew Jewish
language now uh is is quite relevant and
and particularly if you read it from an
active inference perspective and that's
what I have been doing and and as well
with the Alexander technique and it
doing that having that view or that so
it's maybe it's a preference that that
seem things seem to have much more
meaning and uh are much clearer to to be
able to understand um that was just one
point the other thing with about biases
was I was just wondering is um is that
that sort of discussion about bias
within within the um the mechan the
basic mechanics uh we we have
preferences and preferences are what
keep you where you are and for me
preferences are biases within the
Precision mechanics pretty much in there

and because it's an unconscious
inference is is pretty much unconscious
unless we um do something like meditate
uh or pay attention uh investigate then
it remains unconscious but but we
probably have the ability to cognitively
um delve a certain degree there's a
certain penetrability to be able to
that's what Jack of how talks about
penetrability to be able to actually
investigate a little further into our
unconscious inferences um to be able to
begin to understand why we why we think
something but then a step on from there
is that a generative model is made up of
an action perception cycle and I don't
it's just a comment I could be could be
my preference that um a lot of
discussion tends to um in in book club
tends to and even within the broader
literature uh
remain in a sort of a passive State um
in a perceptive state without the
recognition that there's an
inseparability of action and perception
uh which I think is what's at the basis
of active inference we we cannot
separate the two so that with with
anything we think or do it is an a
perception and an action and
um and I think it's
much much more expansive than people's
conceptualizations of how we actually
function in the
world a lot of our actions are UNC
unconscious when their habits they're
habitual when really not present to to
what we're doing meditation and

mindfulness meditation Buddhist
Meditation is is one method of of
beginning to
investigate that act that action
perception Loop that that that enables
us to to function
[Music]
at every minute of the day really as as
well as what I consider the the
Alexander technique uh the Alexander
technique in its true
form uh brings action and perception and
thinking together so that it's based on
the
psychophysical um re-education process
but I I just wanted to put forward I
suppose that that the whole body the
whole body seems to get broken up a bit
in these discussions and and the action
perception cycle cycle is that it's an
intwined uh
Inseparable functioning and I think the
ability for for us to really cognate
that to really sort of understand that
is what what active
inference uh hopefully will and I think
is coming bringing to light it's a thing
that draws people to
it but preferences not biases yeah
one interesting point there to seal that
point is internal and external is just
pick aside but sense and action
are
identically considered by the across any
boundaries like and also I find it
interesting that the Mew for internal
state it is almost like if you rotate
this like clockwise or counterclockwise

180 degrees that it's like it it looks
like the symbol on the other
side to to just show that 's kind of
like a mirroring but but not a an
equivalence but that's the interface
that that enables there to be that
Andrew sure I just wanted to speak on
the I'm a practicing U Buddhist and and
meditate regularly and have had a lot of
thoughts about this although I haven't
become too strongly haven't gone too
deep into the literature on active
inference and meditation but I I very
much agree with this idea of um
meditation is a kind of like practicing
one's
attention um paying attention to one's
thoughts whatever is going on internally
a lot of forms of meditation involved
reading the body and so practicing
interoception and um and with that
there's a kind of I I don't know uh
almost like a an attempt at overcoming
something like habit like I'm getting
back the models like you have your e uh
Matrix versus G and I think by
practicing by being able to have that
sort of meta awareness I suppose around
what your thoughts are what your desires
are in in Buddhist terms or your
preferences as far as the C Matrix goes
and trying to figure out like what what
are those about what am I going for do
you have certain things that you want
that you know you shouldn't want and
like trying to overcome those um yeah I
I think a lot of it involves like
practicing one's attention that is I

guess Precision optimization like trying
to certain um yes and certain signals
that you're receiving and and instead
switching your attention to to others
that might actually help you realize
your your new preferences um as opposed
to maybe whatever like you to the the
negative habit before something and
um but with Buddhism you have this
notion of like a desire being kind of
endless right um and so like it can you
know kind of take you over if uh in the
extreme and I'm not sure where that lies
as far as um you know trying to use um P
mdp to model that as a process or
something goes you know but uh yeah it's
all it's all very very interesting so I
kind of just wanted to chime in on on
that
bit Yeah Yeah like just to connect the
the the references a little bit in the
textbook here was that kind of figure
you alluded to
54 with the Seesaw or the Continuum
between the total pass through of habit
defined however and then the total
sharpening to an arbitrary function
which function that's life SL modeling
but the this arbitrarily defined one
such that any sharpening we can fit
within this framework it doesn't even
have to be one that is created by the
modeler that's why it's a descriptive
view from the
outside and then here in the saned Smith
uh 21 paper live from 28 this kind of
demonstrated the plausibility in the
scheme for nested

awarenesses because just having
attention or bias filter of one depth is
just like sharpens out let's just say
edges from a static image so successive
layers of bias signal the noise could
could pull out edges or or hallucinate
them um but here it kind of pointed
towards the ability that there could
be awareness of awareness and and this
sort of more nuanced view on
phenomenology which could be understood
just could be Whimsical could be
different culture it could be different
individuals just to show that it could
be created in terms of this broader
topic of regimes of
attention attention as internal action
internal action as covert action so
using the same apparatus that's used for
the overt action and like the planning
where to
walk turning that to apply to
potentially sequences of
attentions a
je uh yeah so um I had a different
question not related to sort of
meditation so it's a little different
but I'll go for it okay okay uh I was
just wondering like how
exactly is sort of uh I guess like Free
Will and measurement Independence sort
of how's that related to like the free
energy principle and active inference
like are like do behaviors follow like
certain attractor States at certain
times and does that mean that like when
you're taking a measurement of
something uh the choice between that

measurement is actually
like the choice between that measurement
is statistically
dependent uh with sort
of
um like
you kind of like uh if you're going to
pick you know between A and B was that
choice sort of determined if sort of you
know our decisions themselves follow
sort of this principle of least action
with like the free energy principle and
all of that so yeah I was just wonder
like how is free will ort of formalized
in uh active
inference big question just so that a
few other people can ask here's the
shortest answer look it up in this Koda
there's been a little bit of discussion
on it people have come down hard on one
side or the
other people have put their professional
careers on the line one side or the
other science shows this science shows
that that's how I see that debate
largely at this point so active
inference doesn't actually like tell us
it doesn't lean towards one side as of
yet or it doesn't have a system that
it's based upon other than historically
but it doesn't have a system indicated
so it's it's expressionless about topics
like the relevance of a given thing in a
given situation for a given thing it's
multiple layers of productivity by the
modeler and the the
contextualize before that's even a
complete sentence

so I think it's
just carton horse or something like that
but but understandable and topically
Associated and historically
Associated um but we we should continue
to talk read theoda and write or add
more and obviously it's an ongoing
discussion so I'm not saying it's
permanently one way or the other I'm
just saying that you will find Discord
messages and Kota and transcripts of
people vehemently
believing both things within the last
two
months okay okay yeah um everyone who
wants to how about just a closing round
of thoughts on this like first
discussion on 10 so bronwin then um
Magdalena then anyone
else um just very quickly
um in terms of active influence going
forward you you discussed it a little
bit before in terms of what you know
sort of is the booking 22 what happens
in in the future um I just wonder it
seems to me that from very different
areas in life people get their teeth
they they hear about active inference
and they get their teeth into it and
it's really hard to Let It Go like it's
just like a dog with a bone and um and I
can't see how it it can do anything
other than than evolve and
become um not not denser but sparer in a
way um a much more simple
um Theory to be able to understand H
sent in Behavior I just wonder what
people think about that and what their

their view on the um the future of
active inference is I mean for me it's
something I I haven't been been able to
let go of and it just raises more
questions day after day so yeah I'll
just leave it
there yeah everyone can can give a
closing thought on that or whatever they
want
um yeah my closing thought is
um that uh I first want to
learn active entrance really well I want
to understand it fully I think it has a
lot of promise and
I I every day I do a little bit of
translation into the um scientific
question C I'm working on
um but I feel the same way you do I just
feel like this this is just the
beginning of a very long
journey it's gonna take a lot of work
sorry about
that any any other closing thoughts on
10 or
[Music]
um yeah uh I think active inference is
rather huge in this uh in the context of
chapter 10 presenting the theory as a
broad unified theory I think it's going
to go in many directions for whoever
wants to start modeling in that
direction I myself discovered active
INF like just a kind of hobbyist
interest in psychoanalysis and then I
found uh the work of Mark SS in neuros
psychoanalysis and then one thing led to
another and here I am wanting to read
Carl friston's work uh instead almost um

and and seeing how this is being applied
in feels as diverse as Psychiatry to
robotics so um yeah I think it really
I'm more interested in seeing what kind
of empirical work like what kind of
direct applications um people carry out
um with this Theory and I think that'll
probably end up shaping the field even
more closing
but
yeah
awesome asil do you want to add
anything
yeah today we speak about Daniel denet
and Buddhism I think one thing is common
between Daniel D and Buddhism both of
them deny self uh
in the
meditation um almost you you will lost
self and Daniel Den introdu s is like a
a school um for example after 20 years
in the school everything will be changed
like teacher student bench everything
and
just they have same name and this
General model
uh is I think
exactly whole IG this don't look um
don't suppose
self this active inference make General
model for artificial intelligence and
people braing and
everything generative not uh um with
some um um some abstract idea I think
so thank you I think that's a super
important
point about how it's purely
synthetic and that is being explored

more
recently so however satisfied and
excited we were
previously it continues to
develop and as with all of our
discussions this one probably opens up
more questions than not but these were
all really great points chapter 10's
kind of Epic but it's also like symbolic
and it is where we
jump out you know we're going to close
this meeting we close that book and then
you do something next so that's really
where like breaks the fourth wall breaks
the fifth wall all these like
interesting Dynamics
of our engagement with it and all the
fun things we get to do in textbook
group yeah though we'll have another
discussion next week
earlier then this time in two
weeks we will talk about projects or
future textbook groups right now already
this
form yeah is accurate so you can fill
out the form you can give any other
information
but other than that we're just kind of
chilling cruising down the end of the
year last few scheduled live streams
have a quiet December then come back in
24 anyone's welcome to be a facilitator
for textbook group or we can explore
like other
things so thank you see you next
time bye thank you
Daniel